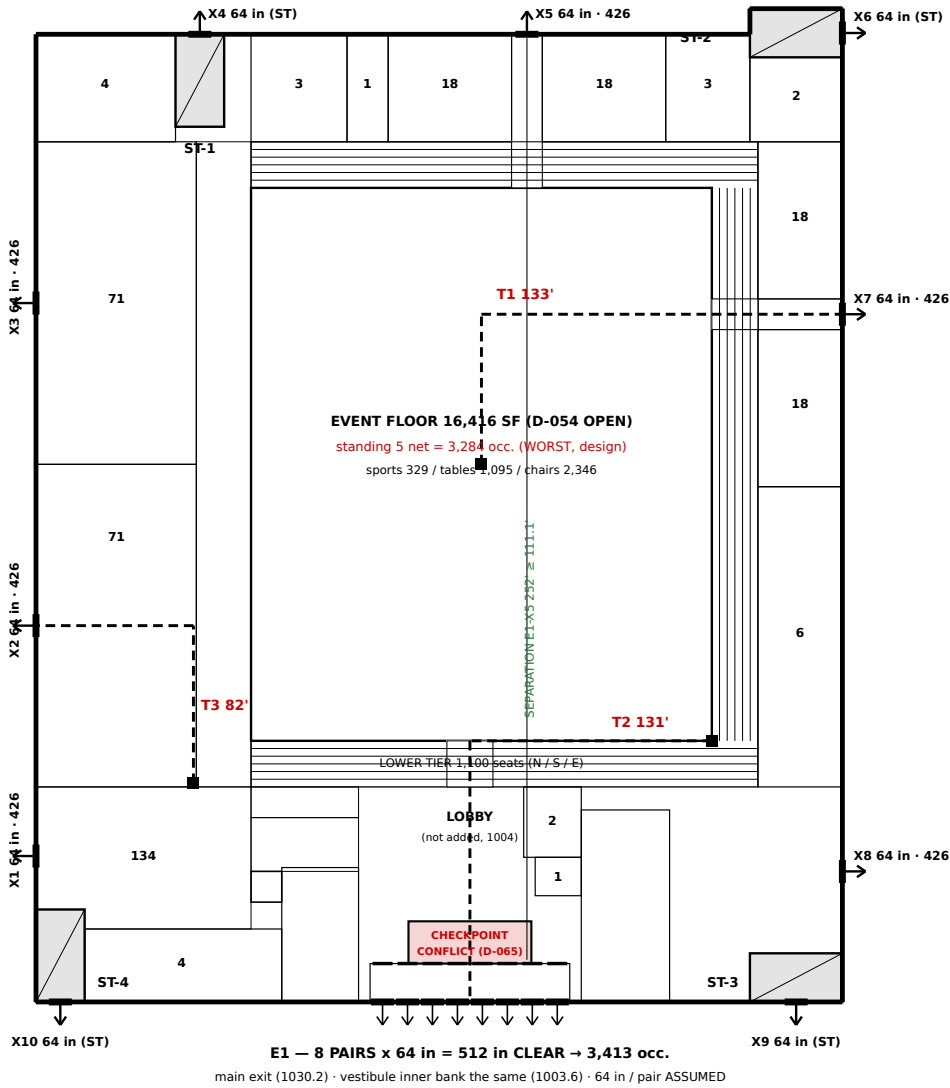


LEVEL 1 — LIFE SAFETY (PLAN REV G)

1" = 50'-0" · north up (approximate) · E1 bank drawn (Option 1 DECIDED)



L1 EXITS — CAPACITY (0.15 in / occ., 1005.3.2 exc.)

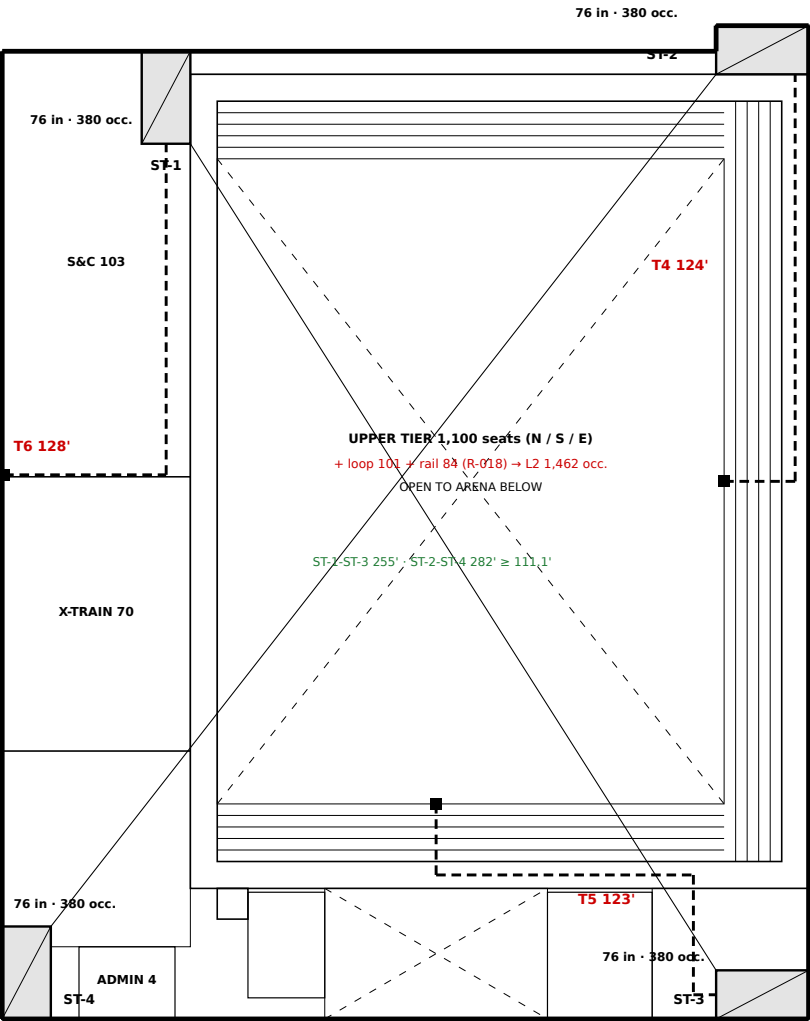
EXIT / CHECK	CLEAR in	CAPACITY	SERVES / RESULT
E1 main (bank)	512	3,413	lobby, portal, south tier, concourses
X1, X2, X3, X5, X7, X8	6 x 64 = 384	426 each	exit-only doors (64 in pair ASSUMED)
X4, X6, X9, X10	4 x 64 = 256	4 x 9.2 in left	stair discharge: L2 uses 54.8 in each
Total, building worst 6,220	need 933.0	have 1,152.0	PASS
Main exit ≥ 1/2 (1030.2)	need 466.5	have 512.0	PASS
Other L1 exits ≥ 1/2 of L1 4,758 (1030.3)	need 356.8	have 420.7	PASS
Lose E1, ≥ 50 % (1005.5)	need 466.5	have 640.0	PASS

TBD

- Exit discharge (1028.3): width from E1 to the public way >= the exit's required capacity, 466.5 in (38.9 ft); the 28 ft Champion Walk / portal opening alone is 336 in — plaza paving beside it must carry the rest; site TBD (D-006, P2-C-101). Access to a public way 1028.5.
- Door widths, hardware and the 10 ft open space / street frontage at E1 (1030.2): architect. Seating aisles, rails, smoke-protected seating (1030.6.2): TBD. Structure 1'-6" ASSUMED (zero margin at 7'-6").

LEVEL 2 — LIFE SAFETY (PLAN REV F, UNCHANGED)

1" = 50'-0" · 4 stairs 76 in (D-052) = the L2 exits · loop = upper concourse on event days



L2 EXITS — STAIRS (0.2 in / occ., 1005.3.1 exc. 1)

EXIT / CHECK	CLEAR in	CAPACITY	SERVES / RESULT
ST-1 ... ST-4	4 x 76 = 304	380 each	one per corner, discharge X4 / X6 / X9 / X10
Total, L2 worst 1,462	need 292.4	have 304.0	PASS (+11.6)
Lose one stair, ≥ 50 % (1005.5)	need 146.2	have 228.0	PASS (+81.8)
Loop at the worst corner (R-018.3, 0.2)	need 78.6	have 84.0	PASS (+5.4)
4 exits from L2 (T1006.3.3, > 1,000)		4 stairs	PASS

Sources: P2-G-002 Rev A; P2-A-101 Rev G / A-102 Rev F (params/phase2_plan_rev_g.yaml); params/phase2_life_safety.yaml; IBC 2021 1003.6, 1005.3, 1005.5, 1006.2.1, 1007.1, 1010.5, 1017.2-3, 1028.3, 1030.2-3, 1030.8 (UpCodes, retrieved 2026-10-04; R-007, R-015, R-018); Shane 2026-10-04 11:07 AM CT.

Schematic life safety diagram for the architect of record and the AHJ. Not a code review and no statement of approval; every line needs architect + AHJ confirmation (D-008). Door widths ASSUMED, travel paths measured on the diagram.

LEGEND

Heavy bar + arrow = exit door (clear in · occupants); E1 = 8 pair bars, outer + inner (vestibule) bank. Grey box = stair. Red dashed + dot = travel path (dot = remote point; feet, rectilinear). Green line = exit separation. Pink zone = checkpoint conflict. Numbers in rooms = occupant load (P2-G-002).

OCCUPANT LOADS (P2-G-002 REV A, IBC T1004.5)

CASE	FLOOR	L1	BLDG	E1 NEED
Sports (50 gross)	329	1,803	3,265	244.9
Tables + chairs (15 net)	1,095	2,569	4,031	302.3
Chairs only (7 net)	2,346	3,820	5,282	396.1
Standing (5 net) (WORST)	3,284	4,758	6,220	466.5
Whole floor zone 17,280 SF standing	3,456	4,930	6,392	479.4

L1 without the floor 1,474 (seats 1,100); L2 1,277 base, 1,462 worst. All cases pass with the 512 in bank; the whole-floor sensitivity still passes (E1 479.4 ≤ 512 in, total 958.8 ≤ 1,152 in).

EXIT SEPARATION (1007.1.1 exc. 2, sprinklered)

Max overall diagonal 333.2 ft (SW corner to NE stair tower) → 1/3 = 111.1 ft (1/2 = 166.6 ft unsprinklered). Measured to any point of a doorway / closest riser (1007.1.1.1).

- L1 E1 ↔ X5: 252.0 ft ≥ 111.1 ft — PASS. E1 measured at the point of the bank (x 87.25-138.75) in line with X5.
- L2 ST-1 ↔ ST-3: 255.1 ft ≥ 111.1 ft — PASS. nearest corners of the stair rectangles (risers lie inside, so the riser distance is not shorter).
- L2 ST-2 ↔ ST-4: 281.5 ft ≥ 111.1 ft — PASS. nearest corners of the stair rectangles.
- Other exits a reasonable distance apart (1007.1.2): one stair per corner, exits on all four walls.

TRAVEL DISTANCE (Table 1017.2: 250 FT)

IBC 2021 Table 1017.2: Group A with sprinklers (903.3.1.1) 250 ft. 1017.3: from the most remote point along the natural and unobstructed path to the entrance to the nearest exit. Paths below are rectilinear along open floor on the diagram (walls / doors / aisles not drawn): UNVERIFIED.

- T1 L1 event floor center → V2 → EXIT (E) → X7: 133 ft — ≤ 250, PASS (UNVERIFIED)
- T2 L1 floor SE corner → portal → lobby → E1: 131 ft — ≤ 250, PASS (UNVERIFIED)
- T3 L1 girls locker far corner → X2: 82 ft — ≤ 250, PASS (UNVERIFIED)
- T4 L2 E upper tier, front row mid → aisle → loop → ST-2: 124 ft — ≤ 250, PASS (UNVERIFIED)
- T5 L2 S upper tier, front row mid → loop → upper concourse (E) → ST-3: 123 ft — ≤ 250, PASS (UNVERIFIED)
- T6 L2 S&C far corner → ST-1: 128 ft — ≤ 250, PASS (UNVERIFIED)

L2 paths end at the stair. If the stairs are open exit access stairways rather than enclosed exit stairways (TBD, architect), travel continues down the stair (1017.3.1, along the nosings) to the L1 exit door: about +50 ft ASSUMED (2 flights + 76 in landings + 5 ft to the door) — still under 250 ft.

COMMON PATH (T1006.2.1, 1030.8)

- Event lockers 1-4 (18 each): one door OK if <= 75 ft: room 32 x 28 / 22 x 41 ft → <= 63 ft rectilinear to any wall point — LIKELY — door not located (UNVERIFIED).
- Boys / girls locker (71 each): > 49 → 2 exits: X2 / X3 + corridor door — OK as planned (corridor door ASSUMED).
- Flex / team assembly (134): > 49 → 2 exits: X1 + lobby side (CD-1 free egress, 1010.2) — OK as planned.
- S&C / cross-training (L2) (103 / 70): > 49 → 2 exit access doorways each (ST-1 / loop; corridor / loop) — ASSUMED doors (TBD).
- Seats (both tiers) (1,100 + 1,100): 30 ft to a choice of two paths: aisle 12-15 ft to the floor / loop, then two ways — TBD — aisles and row lengths not drawn.

Rules: one exit only if ≤ 49 occ. and ≤ 75 ft (T1006.2.1, Group A, sprinklered); seats 30 ft to a choice of two paths (1030.8); unoccupied mech rooms exempt (1006.2.1 exc. 3).

FINDING — SECURITY CHECKPOINT vs EGRESS (OPEN)

Security checkpoint vs egress: the 32 ft checkpoint stands directly inside the E1 bank; clear lobby width beside it = 58 - 32 = 26 ft = 312 in < 466.5 in the main exit needs. Exits must not pass through the checkpoint in the egress direction, and the required width cannot be narrowed (IBC 1003.6, 1010.5, 1030.2).

- Option 1: move screening into a side bay of the lobby (e.g. the west side by CD-1), off the egress path; keep >= 466.5 in (38.9 ft) of clear width from the portal / lobby to the E1 bank, free of lanes, stanchions and queues.
- Option 2: screen outdoors on the entry plaza (temporary lanes beside a clear discharge path >= 466.5 in to the public way); the lobby stays clear. Plaza paving wider than the 28 ft Champion Walk (D-006 site, P2-C-101).

FINDING — RESTROOMS AT THE WORST CASE (OPEN)

Restrooms at the worst case: standing floor + lower seats = 4,384 → 26 men's / 50 women's WC (lav 11 / 15, DF 5) vs 10 / 18 drawn on L1 (P2-G-003 Rev I basis 1,428). Not redesigned.

- Option 1: size the L1 restrooms for the worst case (+16 M / +32 W WC, +7 / +10 lav: about +65 fixtures ≈ +3,250 SF at 50 SF per fixture, phase2_program.yaml ASSUMED).
- Option 2: limit standing / large floor events by a posted occupant load, or bring temporary facilities for them (AHJ acceptance NOT researched).

PRELIMINARY — CONCEPTUAL DESIGN — NOT FOR CONSTRUCTION

PROJECT	PHASE	SHEET TITLE	SCALE	DATE	REV	DRAWN BY	SHEET
Hazel Green Regional Athletic Complex Trojan Horse Arena · Hazel Green area, Madison County, Alabama (site TBD, D-006)	PHASE 2	LIFE SAFETY PLAN LEVELS 1 + 2 · SCHEMATIC	1" = 50'-0"	October 2026	A	Drawn by KEYSTONE (AI) for Shane Brazelton	P2-A-111